



**DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA**

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

**MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE
COMPUTADORES E INFORMÁTICA**

PRACTICAL GUIDE NO. 5

1. Consider two random variables X and Y with the following joint distribution:

$X \setminus Y$	0	1	2
0	0,3	0,2	0
1	0,1	0,15	0,05
2	0	0,1	0,1

- (a) Calculate the marginal probability mass functions of X and Y and plot them.
 (b) Calculate the mean and variance of X and Y .
 (c) Calculate the correlation, covariance, and correlation coefficient between X and Y .
2. Given the joint probability of the variables X and Y shown in the next table, are the variables X and Y independent? Perform the calculations using Matlab.

$X \setminus Y$	-1	0	1
-1	1/8	1/8	1/24
0	1/8	1/4	1/8
1	1/24	1/8	1/24

3. Consider two random variables related to the grades of a class of 120 students in two Curricular Units. Let's designate these discrete variables as $N1$ and $N2$.
- (a) Generate $N1$ and $N2$ so that, before rounding to the nearest integer, they have a Normal distribution with the following means and variances: mean of $N1$ equal to 12; mean of $N2$ 15% greater than $N1$; variances equal to 1/4 of the mean.
 (b) Obtain and represent in a graph the joint probability mass function of $N1$ and $N2$.
 (c) What is the value of the correlation coefficient between $N1$ and $N2$?
 (d) Are $N1$ and $N2$ independent?
4. (Optional) In a given city and at a given time of year, the probability of the weather being sunny is 0,75 and the probability of the weather raining is 0,25.

Local weather forecasts are accurate 100% of the time if it is rainy, but only 1/3 if it is sunny. An attentive student observes that, on average, the meteorologist only has an average accuracy rate of 50%, while if he always predicted good weather (sunny), he would be accurate 75% of the time. Based on this argument, he applies for the position, but the supervisor rejects the application. Explain why? Suggestion: Start by creating the table with the joint probability mass function.